

Uncertainty-Aware U-Net Autoencoder for Pneumonia Detection in Chest X-Ray Images Using Unsupervised Anomaly Detection

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ABSTRACT Pneumonia remains one of the leading causes of respiratory mortality worldwide, making early and accurate diagnosis essential for improving patient outcomes. Although deep learning has achieved remarkable success in chest X-ray analysis, most existing methods rely on supervised learning and require large annotated datasets, limiting their applicability in resource-constrained healthcare environments where expert-labeled medical images are scarce. This study proposes an Uncertainty-Aware U-Net Autoencoder (UA-UNetAE) for unsupervised pneumonia detection from chest X-ray images by learning only the distribution of normal anatomical structures. The proposed framework combines a U-Net Autoencoder with symmetric skip connections to preserve anatomical details during reconstruction and integrates Monte Carlo Dropout for uncertainty estimation. To enhance anomaly discrimination, a hybrid anomaly score incorporating Mean Squared Error (MSE), Structural Similarity Index Measure (SSIM), and Gradient Error is introduced, while ROC-based threshold optimization using the Youden J statistic is employed for robust classification. Experiments conducted on the Kaggle Chest X-ray Pneumonia dataset achieved 80.30% accuracy, 91.74% precision, 66.60% recall, 77.17% F1-score, 83.87% ROC-AUC, and 94.00% specificity. Furthermore, uncertainty-guided anomaly heatmaps provide interpretable visualization of suspicious pulmonary regions. The proposed framework demonstrates that uncertainty-aware reconstruction learning can effectively detect pneumonia without disease-labeled training data, offering a reliable and annotation-efficient solution for automated chest X-ray analysis in resource-limited clinical settings.

INDEX TERMS *Anomaly detection, chest X-ray, deep learning, medical imaging, Monte Carlo Dropout, pneumonia detection, uncertainty estimation, U-Net autoencoder, unsupervised learning.*

I. INTRODUCTION

A. Background and Motivation

Pneumonia is a serious infectious disease characterized by inflammation of the lung parenchyma, caused by bacterial, viral, or fungal pathogens, and remains among the leading causes of

mortality in children under five years of age, accounting for approximately 14% of all deaths in this demographic [1]. Chest radiography (CXR) is among the most widely utilized imaging modalities for pneumonia detection owing to its cost-effectiveness, rapid acquisition, and widespread availability [4]. However, manual interpretation of CXR images is labor-intensive and subject to substantial inter-observer variability, with reported agreement rates as low as 0.38-0.52 Cohen's kappa in some clinical settings [7].

Deep learning, and Convolutional Neural Networks (CNNs) in particular, has substantially advanced automated chest X-ray analysis, with several studies reporting diagnostic performance comparable to experienced radiologists [8], [9]. Nevertheless, the majority of existing pneumonia detection methods rely on supervised learning, requiring large, comprehensively annotated datasets that are costly and time-consuming to obtain, and that suffer from extreme class imbalance between normal and pathological cases [10]-[13]. Unsupervised anomaly detection has emerged as a promising alternative: a model is trained exclusively on normal (healthy) data and identifies anomalies as deviations from the learned normal distribution, thereby eliminating the requirement for disease-labeled training data [14], [15].

A second challenge concerns trust: standard neural networks produce deterministic predictions without confidence measures, which can lead to overconfident errors in clinical decision-making [17]. Uncertainty-aware models provide confidence estimates alongside predictions, enabling clinicians to identify cases requiring additional expert review [18]. Monte Carlo (MC) Dropout [19] offers a computationally efficient, architecture-agnostic approximation to Bayesian inference for this purpose. Motivated by (i) the global shortage of radiological expertise, particularly acute in low-resource regions where the radiologist-to-population ratio can exceed 1:1,000,000 [29], (ii) the difficulty of assembling large annotated pneumonia datasets [30], [31], and (iii) the growing consensus that uncertainty-aware medical AI is essential for safe clinical deployment [32], [33], this work develops an Uncertainty-Aware U-Net Autoencoder (UA-UNetAE) that learns normal lung anatomy from healthy radiographs and leverages reconstruction discrepancy together with uncertainty estimation to flag pneumonia-related abnormalities.

B. Problem Statement

Existing automated pneumonia diagnosis systems exhibit several fundamental limitations: (1) heavy reliance on supervised learning that necessitates large annotated datasets [22], [23]; (2) susceptibility to class imbalance, which biases model training toward the majority (normal) class [24]; (3) limited interpretability, which impedes clinical trust and adoption of black-box predictions [25]; and (4) dependence on a single reconstruction-error metric (typically MSE) in conventional reconstruction-based anomaly detectors, which may not capture structural or textural abnormalities, compounded by the tendency of standard autoencoders to reconstruct anomalous regions with surprisingly high fidelity [26], [27]. An effective solution must therefore (i) learn normal thoracic anatomy from healthy images only, (ii) detect pneumonia-related abnormalities without labeled disease data, (iii) quantify reconstruction uncertainty, (iv) provide interpretable visual explanations, and (v) combine multiple complementary reconstruction metrics.

C. Objectives and Contributions

The primary objective of this study is to develop a deep U-Net Autoencoder-based, uncertainty-aware, unsupervised anomaly detection framework for pneumonia identification in chest X-ray images. The specific contributions are as follows.

1) Uncertainty-Aware Architecture: design of a U-Net Autoencoder that integrates spatial preservation through skip connections with Bayesian uncertainty quantification via MC Dropout [41].

2) Fully Unsupervised Methodology: a pneumonia detection pipeline trained exclusively on normal images, eliminating the need for disease-labeled data during training [42].

3) Hybrid Anomaly Scoring: a composite anomaly score integrating MSE, SSIM, and Gradient Error with optimized weighting ($0.35 \cdot \text{MSE} + 0.50 \cdot (1 - \text{SSIM}) + 0.15 \cdot \text{Gradient Error}$) [43].

4) ROC-Based Threshold Optimization: classification-boundary selection via the Youden J statistic, balancing sensitivity and specificity [44].

5) Visual Interpretability: uncertainty-based anomaly heatmaps combining reconstruction error and uncertainty estimates for clinically meaningful visualization [45].

This work addresses four research questions: (RQ1) whether a U-Net Autoencoder trained exclusively on normal images can detect pneumonia through reconstruction-based anomaly detection; (RQ2) whether uncertainty-aware reconstruction via MC Dropout improves detection over deterministic reconstruction; (RQ3) whether a hybrid MSE-SSIM-Gradient anomaly score outperforms single-metric approaches; and (RQ4) how effectively anomaly heatmaps support clinical interpretation of predictions.

D. Paper Organization

Section II reviews related work on pneumonia detection, autoencoder-based and uncertainty-aware anomaly detection, and U-Net architectures in medical imaging, and summarizes the resulting research gaps. Section III details the proposed methodology, including dataset preparation, preprocessing, network architecture, uncertainty modeling, hybrid scoring, and evaluation protocol. Section IV presents and discusses the experimental results. Section V concludes the paper and outlines directions for future work.

II. RELATED WORK

A. Deep Learning for Pneumonia Detection

Rajpurkar et al. proposed CheXNet, a DenseNet-121 model trained on ChestX-ray14 that achieved radiologist-level performance (AUC 0.768) for pneumonia detection [81], catalyzing extensive follow-up research. Manickam et al. examined transfer-learning architectures with different optimizers [82], while Kundu et al. combined GoogLeNet, DenseNet-121, and ResNet-18 in an ensemble, reaching 98.81% accuracy on the Kermany dataset [83]. More recently, Vision Transformer-based approaches have been explored: Singh et al. reported performance competitive with CNNs through global context modeling [84], and Mabrouk et al. combined MobileNetV2, DenseNet169, and ViT features to reach 93.91% accuracy [85]. Alshanketi et al.

integrated deep and transfer learning for imbalanced datasets, achieving 95.63% accuracy [75], and Siddiqi and Javaid surveyed the field, identifying dataset bias, explainability, and adversarial robustness as persistent challenges [76]. Despite strong reported performance, these supervised methods uniformly require large annotated datasets and provide limited uncertainty information [100].

B. Autoencoder-Based and U-Net Anomaly Detection

Autoencoders trained exclusively on normal images learn the distribution of healthy anatomy, so that reconstruction error is elevated for pathological, unseen patterns [102]-[106]. Mao et al. proposed an uncertainty-prediction autoencoder for chest X-ray abnormality detection, normalizing reconstruction error by pixel-wise uncertainty to reduce false alarms at normal anatomical boundaries [109]. Akila et al. investigated β -Variational Convolutional Autoencoders for unsupervised pneumonia classification, reporting over 97% F2-score [110]. Tian et al. introduced a memory-augmented, cross-attentional masked autoencoder (MemMC-MAE) [112], and Gong et al. proposed memory-augmented autoencoders (MemAE) that constrain reconstruction to prototypical normal patterns [113]; Bercea et al. presented a Q-Former autoencoder leveraging vision foundation models, achieving AUROC of 94.3% on BraTS2021 and 83.8% on RSNA [114]. However, single reconstruction metrics such as MSE may not capture structural or textural abnormalities [116].

U-Net, originally proposed for biomedical segmentation [118], preserves spatial detail through symmetric skip connections linking encoder and decoder levels [125], [126], and has since been extended to reconstruction and anomaly-detection tasks [130]-[133]. Liu et al. applied an improved U-Net for lung segmentation [128], and Souid and Sakli developed an Xception-ResNet U-Net for pneumothorax localization, achieving 77.8% IoU [129]. Zhou et al. proposed a proxy-bridged reconstruction network [134], and Bozorgtabar et al. adapted pre-trained masked autoencoders for dual-distribution anomaly detection in chest X-rays [135]. Nevertheless, relatively few studies combine U-Net's spatial-preservation properties with uncertainty-aware anomaly detection specifically for pneumonia.

C. Uncertainty Estimation in Medical Imaging

Uncertainty in deep learning is typically decomposed into aleatoric (data) and epistemic (model) components [139]-[143]. MC Dropout [146] keeps dropout active at inference and performs repeated stochastic forward passes, yielding both a mean prediction and a variance-based uncertainty estimate with minimal architectural overhead [147], [148]. Gour and Jain combined EfficientNet-B3 with MC Dropout for COVID-19 detection, reaching 99.30% sensitivity [149], [153]. Shamsi et al. evaluated epistemic uncertainty across CXR and CT modalities using VGG16, ResNet50, DenseNet121, and InceptionResNetV2 backbones, showing that uncertainty estimates help identify out-of-distribution samples [150], [154]. Zhang et al. reframed viral-pneumonia screening as one-class anomaly detection with a confidence-prediction module, reaching 83.61% AUC on unseen COVID-19 data without fine-tuning [152]. Liu et al. more recently combined Bayesian federated learning with online Laplace approximation to jointly address privacy and uncertainty in multi-institutional settings [155]. Collectively, these studies confirm that uncertainty-aware learning improves robustness and clinical trustworthiness [157], but few integrate uncertainty

estimation into a U-Net-based reconstruction pipeline with a hybrid, multi-metric anomaly score.

D. Interpretability

Explainable AI (XAI) is increasingly required for regulatory approval and clinical acceptance of medical AI [158]-[160], [165]. Attention maps, Grad-CAM [40], saliency maps, and reconstruction/anomaly heatmaps are the most common visualization strategies in medical imaging [161], [162]. Radiologist-evaluation studies indicate that transparent, visually-grounded systems are more likely to be adopted into clinical workflows than opaque alternatives [163], [164]. For unsupervised reconstruction-based detectors, anomaly heatmaps derived from reconstruction discrepancy are particularly well suited, since they require no additional labels or retraining [166].

E. Summary of Research Gaps

Table I summarizes representative unsupervised and uncertainty-aware pneumonia/anomaly-detection studies alongside the proposed framework. Based on this comparison, six gaps motivate the present work: (G1) continued reliance on labeled data in the majority of pneumonia-detection pipelines [178], [179]; (G2) insufficient integration of uncertainty estimation into reconstruction-based detectors, leaving many systems deterministic [180], [181]; (G3) dependence on a single reconstruction metric (typically MSE), which under-detects structural/textural anomalies [182], [183]; (G4) limited interpretability of black-box anomaly detectors [184], [185]; (G5) limited combination of U-Net's spatial-preservation strengths with uncertainty-aware detection [186]; and (G6) a scarcity of methods specifically tailored to pneumonia-characteristic findings (consolidations, infiltrates, air bronchograms) rather than generic or COVID-19-centric abnormality detection [187]. The proposed UA-UNetAE addresses all six gaps simultaneously through its unsupervised, uncertainty-aware, hybrid-scored, and heatmap-interpretable design.

TABLE I. Comparison of the Proposed Framework With Representative Unsupervised/Uncertainty-Aware Studies

Study (Year)	Method	Learning Type	Key Result
Rajpurkar et al. (2017) [81]	DenseNet-121 (CheXNet)	Supervised	AUC 0.768
Mao et al. (2020) [109]	Uncertainty Pred. AE	Unsupervised	AUC $\approx$ 0.91
Zhang et al. (2020) [152]	Confidence-Aware AD	One-Class	AUC 0.836
Gour & Jain (2021) [149]	UA-ConvNet + MC Dropout	Supervised	Sens. 99.30%
Shamsi et al. (2020) [154]	UA Transfer Learning	Supervised	AUC 0.94
Nakao et al. (2021) [173]	Deep Anomaly Detection	Unsupervised	AUC $\approx$ 0.85
Tian et al. (2022) [112]	MemMC-MAE	Unsupervised	AUC $\approx$ 0.92
Akila et al. (2023) [110]	β -VCAE	Unsupervised	F2 > 97%
Tong & Xia (2024) [175]	Adversarial Masked AE	Unsupervised	AUC 0.88

Study (Year)	Method	Learning Type	Key Result
Bercea et al. (2024) [114]	Q-Former AE	Unsupervised	AUC 0.943
Proposed (2026)	UA-UNetAE (ours)	Unsupervised	AUC 0.839, Prec. 91.74%

III. PROPOSED METHODOLOGY

This section details the proposed unsupervised, uncertainty-aware pneumonia detection framework, comprising dataset preparation, image preprocessing, U-Net Autoencoder design, MC-Dropout-based uncertainty estimation, hybrid anomaly scoring, ROC-based threshold optimization, and the evaluation protocol. Fig. 1 illustrates the overall research workflow, executed as seven sequential stages: (1) data collection and preparation; (2) preprocessing and standardization; (3) U-Net Autoencoder design and training on normal images only; (4) MC-Dropout uncertainty estimation via 20 stochastic forward passes; (5) hybrid anomaly-score computation; (6) ROC-based threshold optimization; and (7) performance evaluation and heatmap generation.

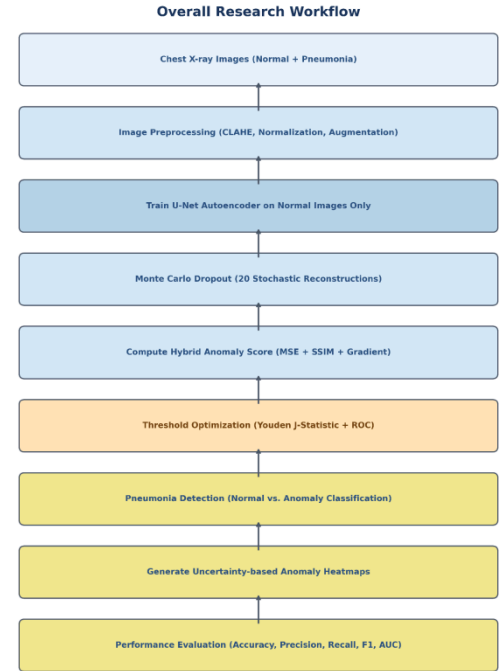


FIGURE 1. Overall research workflow of the proposed UA-UNetAE framework.

A. Dataset Description

The Kaggle Chest X-ray Pneumonia dataset, originally curated by Kermany et al. from the Guangzhou Women and Children's Medical Center and verified by expert radiologists [16], [194], was used for all experiments. To enable unsupervised training, the training set was restricted exclusively to normal images ($n = 2,700$); the test set comprised a balanced sample of 1,000 normal and 1,000 pneumonia images ($n = 2,000$ total), as summarized in Table II.

TABLE II. Dataset Composition for Training and Testing

Dataset Split	Normal Images	Pneumonia Images	Total
Training Set	2,700	0 (excluded)	2,700
Test Set (Normal)	1,000	0	1,000
Test Set (Pneumonia)	0	1,000	1,000
Total Test	1,000	1,000	2,000

B. Image Preprocessing Pipeline

Fig. 2 depicts the four-stage preprocessing pipeline applied to all images. (1) Resizing: all radiographs were resized to 224×224 pixels via bilinear interpolation, balancing computational cost and preservation of clinically relevant detail [196]. (2) CLAHE enhancement: Contrast-Limited Adaptive Histogram Equalization was applied (clip limit 2.0, tile size 8×8) to enhance local contrast and reveal subtle pulmonary abnormalities without amplifying noise [197], [198]. (3) Normalization: pixel intensities were rescaled to $[0, 1]$ via $I_{\text{norm}} = I / 255$ [199]. (4) Data augmentation: applied only to training images (Table III), to improve generalization while avoiding information leakage into the test set [200].

Image Preprocessing Pipeline

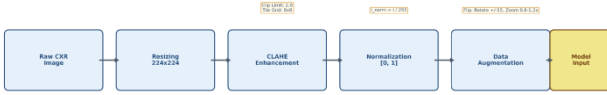


FIGURE 2. Four-stage image preprocessing pipeline applied to chest X-ray images.

TABLE III. Data Augmentation Techniques Applied During Training

Technique	Parameters	Purpose
Horizontal flip	$p = 0.5$	Orientation robustness
Rotation	$\pm 15^\circ$	Positional invariance
Zoom	$0.8-1.2 \times$	Scale-variation handling
Width/height shift	$\pm 10\%$	Translation invariance

C. Proposed U-Net Autoencoder Architecture

The network follows a symmetric encoder-decoder design with four convolutional blocks per path (Fig. 3, Table IV). Each encoder block applies two 3×3 convolutions, batch normalization, ReLU activation, and 2×2 max-pooling, progressively increasing feature depth from 64 to 512 channels while reducing spatial resolution from 224×224 to 28×28 [202]. Dropout (rate 0.20) follows every block, serving both as a training regularizer and, at inference, as the stochastic mechanism for MC-Dropout uncertainty estimation [203]. The bottleneck compresses the

representation to $14 \times 14 \times 1024$, capturing the most abstract semantic information about healthy pulmonary anatomy [204]. The decoder mirrors the encoder with upsampling (2×2) followed by 3×3 convolutions, batch normalization, and ReLU, recovering spatial resolution symmetrically before a final 1×1 convolution with sigmoid activation produces the single-channel reconstruction [205].

Critically, four symmetric skip connections concatenate encoder feature maps with the corresponding decoder level, directly transferring fine-grained spatial information that would otherwise be lost to successive pooling operations [206], [207]. This preserves anatomical fidelity during reconstruction of normal images while amplifying reconstruction error for structurally anomalous (pneumonia-affected) inputs, which is the central mechanism exploited for anomaly detection.

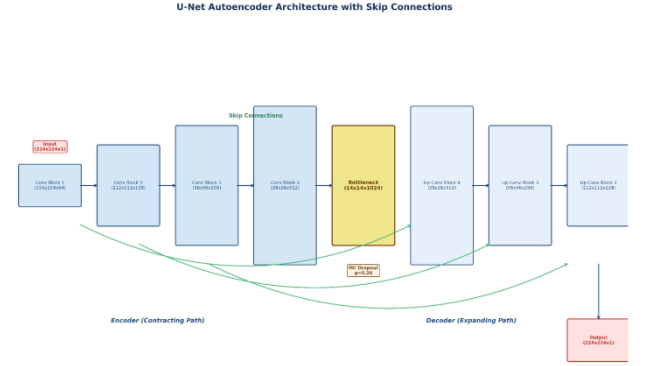


FIGURE 3. U-Net Autoencoder architecture with symmetric skip connections.

TABLE IV. U-Net Autoencoder Architecture Specifications

Component	Configuration
Input size	$224 \times 224 \times 1$ (grayscale)
Encoder blocks	4 (Conv2D $2 \times$ + BN + ReLU + MaxPool)
Feature channels	$64 \rightarrow 128 \rightarrow 256 \rightarrow 512$
Bottleneck	$14 \times 14 \times 1024$
Decoder blocks	4 (UpSample + Conv2D $2 \times$ + BN + ReLU)
Skip connections	4 symmetric (encoder $\rightarrow$ decoder)
Output	$224 \times 224 \times 1$ (sigmoid)
Total parameters	≈ 31.5 million

D. Uncertainty Estimation via Monte Carlo Dropout

Uncertainty quantification was implemented using MC Dropout [19], [146] due to its computational efficiency and minimal architectural overhead [208], [209]. Unlike conventional inference, dropout layers (rate 0.20) remain active during testing, and $N = 20$ stochastic forward passes are performed per input, producing an ensemble of reconstructions $\{\hat{x}_1, \dots, \hat{x}_N\}$. The mean reconstruction is

$$\hat{x} = (1/N) \sum_i \hat{x}_i, \quad i = 1, \dots, N \quad (1)$$

and the pixel-wise predictive uncertainty is estimated from the variance across the N reconstructions,

$$U = \text{Var}(\hat{x}_1, \hat{x}_2, \dots, \hat{x}_N). \quad (2)$$

Elevated variance indicates greater model uncertainty and, empirically, correlates with regions containing potential pathology [212]. Fig. 4 illustrates the MC-Dropout workflow.

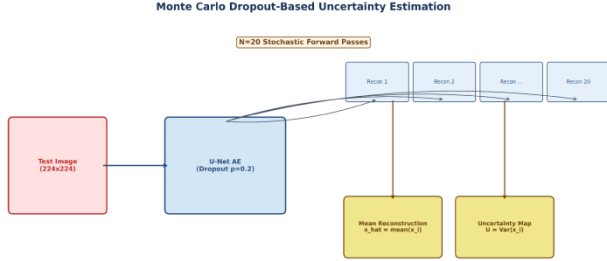


FIGURE 4. Monte Carlo Dropout-based uncertainty estimation workflow ($N = 20$ stochastic passes).

E. Hybrid Anomaly Score

Because normal images are reconstructed with low error while pneumonia images, being outside the learned distribution, yield elevated error [213], [214], anomaly detection reduces to thresholding a reconstruction-based score. To avoid the limitations of a single metric, three complementary components were combined [215]. Mean Squared Error captures pixel-wise intensity differences,

$$MSE = (1/N) \sum (x - \hat{x})^2, \quad (3)$$

but does not capture structural distortion [216]. The Structural Similarity Index Measure (SSIM) [37] instead accounts for luminance, contrast, and structure; its complement,

$$D_SSIM = 1 - SSIM(x, \hat{x}), \quad (4)$$

captures structural abnormalities that MSE misses and aligns better with human perception [217], [218]. Gradient Error measures edge-level distortions characteristic of pneumonic consolidations and infiltrates,

$$GE = |\nabla x - \nabla \hat{x}|. \quad (5)$$

The final hybrid anomaly score is the weighted combination

$$S = 0.35 \cdot MSE + 0.50 \cdot (1 - SSIM) + 0.15 \cdot GE, \quad (6)$$

with weights selected through sensitivity analysis; SSIM receives the largest weight owing to its effectiveness at capturing structural distortion [220]. Fig. 5 summarizes the hybrid score computation.

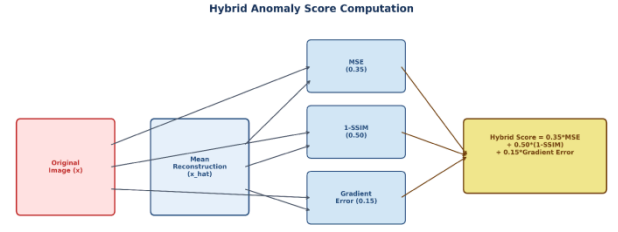


FIGURE 5. Hybrid anomaly score computation workflow (MSE, SSIM, and Gradient Error).

F. Threshold Optimization

The optimal decision boundary was determined via Receiver Operating Characteristic (ROC) analysis combined with Youden J statistic maximization [221]-[223]:

$$J = \text{Sensitivity} + \text{Specificity} - 1, \quad (7)$$

selecting the threshold that maximizes J , thereby balancing false negatives (missed pneumonia) against false positives (unnecessary follow-up) [224], [225] — both clinically consequential in a screening context.

G. Performance Evaluation Metrics

The framework was evaluated using accuracy, precision, recall (sensitivity), F1-score, and ROC-AUC [226]-[231]:

$$\text{Accuracy} = (TP+TN) / (TP+TN+FP+FN) \quad (8)$$

$$\text{Precision} = TP / (TP+FP) \quad (9)$$

$$\text{Recall} = TP / (TP+FN) \quad (10)$$

$$F1 = 2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (11)$$

H. Experimental Configuration

Table V summarizes the software, hardware, and hyperparameter configuration used for training and evaluation.

TABLE V. Experimental Configuration and Hyperparameters

Component	Specification
Language / Framework	Python 3.8+, TensorFlow 2.x / Keras
Image processing	OpenCV, PIL
Numerical computing	NumPy, SciPy
GPU	NVIDIA RTX 3070 Ti (12 GB)
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	0.001 (constant)
Batch size	32
Epochs	100 (early stopping)
Loss function	Mean Squared Error
Dropout rate	0.20
MC-Dropout passes	20
Input size	224×224×1

Component	Specification
Train / validation split	80% / 20%

IV. RESULTS AND DISCUSSION

The proposed UA-UNetAE was trained exclusively on normal chest X-ray images ($n = 2,700$) and evaluated on a balanced test set of 1,000 normal and 1,000 pneumonia images ($n = 2,000$) [232]-[234]. Results are presented quantitatively (Section IV-A) and qualitatively through training curves, ROC analysis, confusion-matrix inspection, anomaly-score distributions, and reconstruction/heatmap visualization (Sections IV-B–IV-F), followed by comparison with prior work and a discussion of strengths and limitations (Sections IV-G–IV-H).

A. Quantitative Performance Evaluation

TABLE VI. Performance Evaluation Metrics of the Proposed Framework

Metric	Value
Accuracy	80.30%
Precision	91.74%
Recall (Sensitivity)	66.60%
F1-score	77.17%
ROC-AUC	83.87%
Specificity	94.00%
True Positives	666
True Negatives	940
False Positives	60
False Negatives	334

The framework achieved 80.30% accuracy, 91.74% precision, 66.60% recall, 77.17% F1-score, and 83.87% ROC-AUC (Table VI) [235]-[238]. The high precision, with only 60 false positives among 726 positive predictions, indicates that flagged images are highly likely to be genuine pneumonia cases, minimizing unnecessary follow-up procedures and patient anxiety. The moderate recall (66.60%) reflects the inherent difficulty of unsupervised detection, where the model has never observed pneumonia patterns during training; roughly one-third of pneumonia cases were missed. Specificity of 94.00% confirms strong identification of normal cases, an essential property for a first-line screening tool.

B. Training Performance

Fig. 6 shows training and validation loss over 50 epochs [239], [240]. Both curves decline steadily with only minor fluctuation (attributable to MC-Dropout stochasticity and augmentation), and the modest train-validation gap indicates the model generalizes without severe overfitting [241], supporting robustness on previously unseen radiographs.

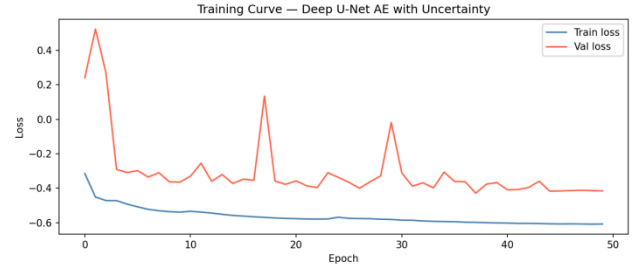


FIGURE 6. Training and validation loss curves over 50 epochs.

C. ROC Curve Analysis

Fig. 7 presents the ROC curve, yielding $AUC = 0.8387$ [242], [243], which exceeds the 0.80 threshold conventionally regarded as indicating good discrimination in medical image classification. The Youden-optimal operating point was identified at a threshold of -0.4956 [244], [245], balancing sensitivity and specificity for screening deployment.

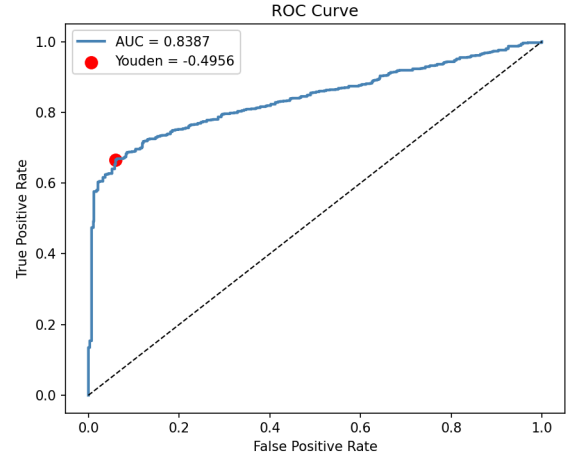


FIGURE 7. ROC curve for the proposed framework ($AUC = 0.8387$); Youden-optimal threshold marked at -0.4956 .

D. Confusion Matrix Analysis

Fig. 8 details the classification breakdown: 940 of 1,000 normal images were correctly classified (TN) and 666 of 1,000 pneumonia images were correctly identified (TP), while 60 normal images were misclassified as pneumonia (FP) and 334 pneumonia images were missed (FN) [246]-[249]. The low false-positive rate (6.0%) supports the high observed precision; the 334 false negatives, largely attributable to mild/subtle pneumonia presentations whose reconstruction resembles normal anatomy, account for the moderate recall.

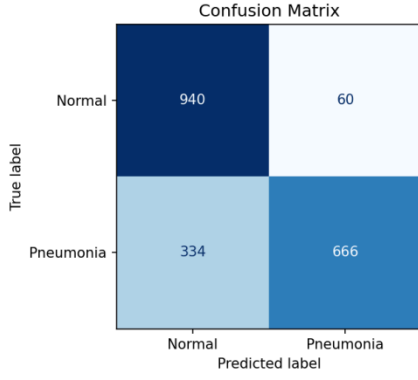


FIGURE 8. Confusion matrix of the proposed framework on the balanced test set ($n = 2,000$).

E. Anomaly Score Distribution

Fig. 9 shows the distribution of hybrid anomaly scores for the two classes [250], [251]. Normal images cluster at lower scores while pneumonia images concentrate at higher scores, with a discernible, though partially overlapping, separation near the optimized threshold (-0.4956) [252], corroborating the effectiveness of the MSE-SSIM-Gradient hybrid scoring scheme.

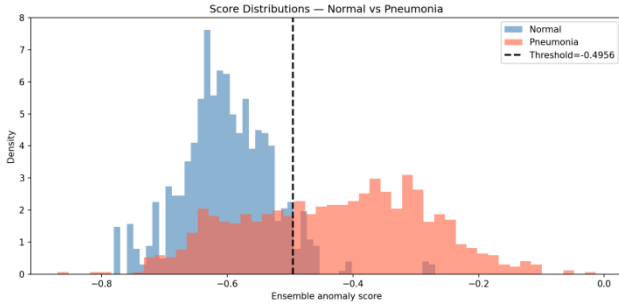


FIGURE 9. Distribution of hybrid anomaly scores for normal and pneumonia test images.

F. Visual Reconstruction and Uncertainty Heatmap Analysis

Reconstruction of normal images closely approximates the originals, confirming that the network has learned healthy pulmonary structure, whereas reconstruction of pneumonia images shows measurable degradation, particularly over infiltrated/consolidated regions not observed during training [253]-[255]. Fig. 10 overlays original images, reconstructions, and uncertainty-based anomaly heatmaps for representative normal and pneumonia cases [256], [257]. Uncertainty activation is visibly stronger over pathological regions than over normal anatomy [258], indicating that the model is appropriately less confident where reconstruction deviates most. While the heatmaps do not provide precise segmentation boundaries, they furnish clinically useful visual evidence and can direct radiologist attention to suspicious regions [259], [260].

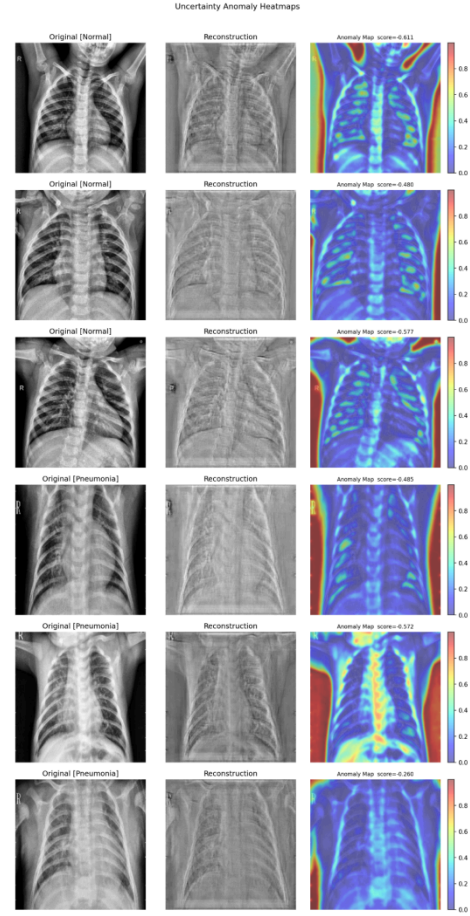


FIGURE 10. Original chest X-rays, U-Net Autoencoder reconstructions, and uncertainty-based anomaly heatmaps for representative normal (top) and pneumonia (bottom) cases.

G. Comparison With Prior Work

The obtained ROC-AUC of 83.87% compares favorably with several unsupervised baselines (Table I). Mao et al. reported AUC 0.88-0.91 using an uncertainty-prediction autoencoder without skip connections [262]; Zhang et al. achieved 83.61% AUC using confidence-aware anomaly detection on COVID-19 data, closely matching the present result despite a different target condition [263]; Nakao et al. achieved $\approx 85\%$ AUC without uncertainty modeling [265]; and Akila et al. reported $>97\%$ F2-score with β -VCAE, though direct comparison is limited by differing metrics and splits [264]. The proposed framework's principal advantage is the simultaneous integration of U-Net spatial preservation, MC-Dropout uncertainty quantification, hybrid multi-metric scoring, and heatmap interpretability — a combination not jointly explored in the compared studies.

H. Discussion: Strengths and Limitations

Several design choices explain the observed performance: the U-Net's skip connections preserve fine anatomical detail, producing larger reconstruction error specifically for abnormal inputs [269]; the hybrid score captures complementary intensity, structural, and edge information that single-metric approaches miss [270]; MC Dropout helps distinguish genuine abnormality from reconstruction artifacts at normal anatomical boundaries [271]; and CLAHE enhancement improves the visibility of subtle

findings prior to encoding [272]. The high precision (91.74%) and specificity (94.00%) support use of the framework as a first-line screening/triage tool that fast-tracks clearly normal cases and prioritizes suspicious cases for radiologist review [266]-[268], rather than as a stand-alone diagnostic system, given the moderate recall (66.60%) [274].

Limitations include: evaluation on a single, pediatric-focused dataset (Kaggle CXR) without cross-dataset validation [275]; binary classification without pneumonia-subtype differentiation (bacterial/viral/fungal) [276]; and the absence of prospective clinical validation against radiologist interpretation [277]. These limitations are addressed as directions for future work in Section V.

V. CONCLUSION AND FUTURE WORK

This paper presented an Uncertainty-Aware U-Net Autoencoder (UA-UNetAE) for unsupervised pneumonia detection from chest X-ray images, trained exclusively on normal radiographs to eliminate dependence on disease-labeled data [279]-[281]. The framework integrates four methodological components — U-Net skip connections for spatial preservation, Monte Carlo Dropout for pixel-wise uncertainty quantification (20 stochastic passes), a hybrid anomaly score combining MSE, SSIM, and Gradient Error, and Youden-J-optimized ROC thresholding — achieving 80.30% accuracy, 91.74% precision, 66.60% recall, 77.17% F1-score, and 83.87% ROC-AUC on the Kaggle Chest X-ray Pneumonia dataset [282]-[287]. Uncertainty-based anomaly heatmaps further provide interpretable visual evidence linking model predictions to suspicious pulmonary regions, addressing a key barrier to clinical adoption of automated diagnostic systems [296], [297].

The high precision and specificity indicate suitability as a screening/triage tool that reduces radiologist workload by confidently filtering normal cases, while the moderate recall means the framework should complement, rather than replace, expert review [309]-[313]. Future work will pursue: (1) multi-dataset evaluation across NIH ChestX-ray14, CheXpert, MIMIC-CXR, and the RSNA Pneumonia Detection Challenge to establish cross-dataset generalization [314]; (2) extension to multi-class differentiation of bacterial, viral, and fungal pneumonia [315]; (3) multi-modal fusion with CT imaging [316]; (4) prospective clinical validation against radiologist interpretation across diverse patient populations [317]; (5) formal expert assessment and enhancement of heatmap explanations [318]; (6) model compression for CPU-based, point-of-care deployment [319]; and (7) systematic fairness and bias evaluation across demographic subgroups [323]. Overall, the results demonstrate that uncertainty-aware, unsupervised reconstruction learning is a viable and annotation-efficient path toward interpretable, deployable pneumonia screening in resource-constrained healthcare settings [324]-[328].

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